BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Compactness in Metric Spaces

Heine-Borel Theorem and Extreme Value Theorem

April 24, 2026

Conducted by

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

Open Cover and Subcover

Open Cover

Let A be a subset of a metric space (X, d). An *open cover* of A is a collection $\{U_\alpha\}_{\alpha \in I}$ of open subsets of X such that

$$A \subseteq \bigcup_{\alpha \in I} U_\alpha.$$

Subcover

A *subcover* of an open cover $\{U_\alpha\}_{\alpha\in I}$ of a set A is a subcollection $\{U_\alpha\}_{\alpha\in J}$, where $J\subseteq I$, such that

$$A\subseteq\bigcup_{\alpha\in J}U_\alpha.$$

Finite Subcover

A *finite subcover* of an open cover $\{U_\alpha\}_{\alpha \in I}$ of a set A is a subcollection $\{U_\alpha\}_{\alpha \in J}$, where $J \subseteq I$ is a finite set, such that

$$A \subseteq \bigcup_{\alpha \in J} U_\alpha.$$

Compactness

Compactness

A subset A of a metric space (X, d) is called *compact* if every open cover of A has a finite subcover. That is, if $\{U_\alpha\}_{\alpha \in I}$ is an open cover of A, then there exists a finite subset $J \subseteq I$ such that

$$A \subseteq \bigcup_{\alpha \in J} U_\alpha.$$

Problem

Show that the Real line $\mathbb{R}$ is not compact in the standard metric.

Problem

Show that the closed interval $(0, 1)$ is not compact in the standard metric on $\mathbb{R}$.

Problem

Show that any finite subset of real numbers is compact in the standard metric on $\mathbb{R}$.

Problem

Show that the closed interval $[0, 1]$ is compact in the standard metric on $\mathbb{R}$.

Why Compactness is Important?

Boundness of Functions

Boundedness of Continuous Functions on Compact Sets

Let (X, d) be a metric space and let $f : X \to \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then f is bounded.

Some Important Theorems on Compactness

Theorem

Every closed subset of a compact metric space is compact.

Theorem

Every compact subset of a metric space (X, d) is closed.

Theorem

Every compact subset of a metric space (X, d) is bounded.

Heine-Borel Theorem

Heine-Borel Theorem

A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.

Continuous Image and Extreme Value Theorem

Continuous Image of Compact Set is Compact

Let (X, d) be a metric space and let $f : X \to \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then $f(K)$ is compact in $\mathbb{R}$.

Extreme Value Theorem

Extreme Value Theorem

Let (X, d) be a metric space and let $f : X \to \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then f attains its maximum and minimum values on K.

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

Scan for the channel

References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.